Mutually Independent Hamiltonian Cycles of $C_k \times C_{k+1}$

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Abstract

In a graph G, a cycle is Hamiltonian cycle if it contains all vertices of G exactly once. Two Hamiltonian cycles $C_1 = \langle u_0, u_1, u_2, ..., u_{n-1}, u_0\rangle$ and $C_2 = \langle v_0, v_1, v_2, ..., v_{n-1}, v_0\rangle$ in G are called independent if $u_0 = v_0$ and $u_i \neq v_i$ for all $1 \leq i \leq n-1$. A set of Hamiltonian cycles $\{C_1, C_2, ..., C_k\}$ of G are called mutually independent if any two different Hamiltonian cycles in $\{C_1, C_2, ..., C_k\}$ are independent. The mutually independent Hamiltonianicity of graph G, IHC(G), is the maximum integer k such that for any vertex u of G there exist k-mutually independent Hamiltonian cycles of G starting at u.

In this paper, we prove that for any positive integer $k \geq 3$, IHC($C_k \times C_{k+1}$) = 4.

1 Introduction

Interconnection network is useful in many areas of physic life, therefore how to design an interconnection network with efficient performance be an important problem. When designing an interconnection network, we often transform this problem to graph for discussing and studying. Formally, a processor is transformed to a vertex and the connection between two processors is transformed to be an edge. By this transformation, an interconnection network can be transformed to a graph.

Cycle is the fundamental class of network topologies, it is a popular network topology for designing parallel and distributed computing algorithm [6,16,17]. Consider networks A and B, if network A can be embedded on network B, then all algorithms for network A can be automatically implemented on network B. For example, if a network can embed cycles, so that the algorithms designed on cycles can be executed on the embed cycles. By this transformation, these algorithms which designed on cycle can be executed on the network.

For definitions and notations, we follow [3]. A (*simple*) *graph* $G = (V, E)$, V, E are two finite sets, where V is called *vertex* set and E is called *edge* set, which defined as V is an nonempty set and E is 2-element subsets of V. $V(G)$ and $E(G)$ denotes *vertex set* and *edge set* of G, respectively. For $u, v \in V(G)$, if $e = uv$ is an edge of a graph G, then we can say that u and v are *adjacent*. A *path* $P[v_0, v_k] = \langle v_0, v_1, ..., v_k\rangle$ in G, defined as a sequence of adjacent vertices and for all $0 \leq i < j \leq k, v_i \neq v_j$. Let $P(i) = v_{i-1}$ be the ith vertex of path P for $1 \leq i \leq k+1$. A *cycle* $C = \langle v_0, v_1, ..., v_k, v_0\rangle$, defined as a sequence of adjacent vertices and for all $0 \leq i < j \leq k, v_i \neq v_j$. Let $C(i) = v_{i-1}$ be the ith vertex of cycle C for $1 \leq i \leq k+1$. In G, a cycle is called *Hamiltonian cycle* if it contains all vertices of G. If there exists a Hamiltonian cycle in graph G, then G is *Hamiltonian graph*. Two Hamiltonian cycles $C_1 = \langle u_0, u_1, u_2, ..., u_{n-1}, u_0\rangle$ and $C_2 = \langle v_0, v_1, v_2, ..., v_{n-1}, v_0\rangle$ in G are *independent* if $u_0 = v_0, u_i \neq v_i$ for all $1 \leq i \leq n-1$. A set of Hamiltonian cycles $\{C_1, C_2, ..., C_k\}$ of G are *mutually independent* if any two different Hamiltonian cycles of $\{C_1, C_2, ..., C_k\}$ are independent. The *mutually independent Hamiltonianicity* of graph G, IHC(G), is the maximum integer k such that for any vertex u of G there exist k-mutually independent Hamiltonian cycles of G starting at u.

A graph is called *cycle*, denoted by C_n, if $V(C_n) = \{v_0, v_1, ..., v_{n-1}\}$ and $E(C_n) = \{v_i v_j$: for $0 \leq i \leq n-2, j = i+1$ and $i = 0, j = n-1\}$. The *Cartesian product* of graphs G and H, written by $G \times H$, is the graph with vertex set $V(G) \times V(H)$ specified by putting (u, v) adjacent to (u', v') if and only if (1) $u = u'$ and $vv' \in E(H)$, or (2) $v = v'$ and $uu' \in E(G)$. Two graphs G_1 and G_2 are *isomorphic* if there is a one-to-one function ϕ from $V(G_1)$ onto $V(G_2)$ such that $uv \in E(G_1)$ if and only if $\phi(u)\phi(v) \in$

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$E(G_2)$. For a vertex $u \in V(G)$, the *neighbor* of u, $N_G(u)$, is the set $\{v : (u,v) \in E(G)\}$. And $\deg_G(u) = |N_G(u)|$ is called the *degree* of u. In a graph G, we use $\Delta(G)$ and $\delta(G)$ to denote the *maximum degree* and *minimum degree* of G such that $\Delta(G) = \max\{\deg_G(u)|u \in V(G)\}$, $\delta(G) = \min\{\deg_G(v)|v \in V(G)\}$. When we consider IHC($G$), these mutually independent Hamiltonian cycles are starting at any vertex of G, then we can get a property as follows:

Property 1 *For any graphs G_1 and G_2, IHC$(G_1 \times G_2) \leq \delta(G_1 \times G_2) = \delta(G_1) + \delta(G_2)$.*

The concept of the mutually independent Hamiltonian cycles can be applied in many different areas. For example, communication applications on the interconnection network are often viewed as the informations interchanged between local computation and global communication stages. Such applications usually be used in transmitting packets equal size along the communication links. In an interconnection network, if we hope to transmitted packets to all processors, but we do not want that some processors take two or more packets in the same time. This kind of problems can be solved by using the concept of mutually independent Hamiltonian cycles.

Mutually independent Hamiltonian cycle problem is an important property for graph theory, there are many studies for this problem on different graph [2,4,5,7,8,10,11]. For hypercube, it has been studied in [12]. For the pancake graph and the star graph, there is a discussing on it [9]. In 2011, we study this problem on $C_n \times C_n$ for n is positive integer [14]. For k-ary n-cubes when k is even [11], it is discussed in the same year. In 2012, we discuss this problem on $C_m \times C_n$ for m and n are both odd [15].

Since $C_k \times C_{k+1}$ and $C_{k+1} \times C_k$ are isomorphic, without loss of generality, we assume that m is even, n is odd and $|m-n| = 1$. By this definitions, we know that graph $C_k \times C_{k+1}$ can be considered as $C_m \times C_n$ for $|m-n| = 1$. In this paper, we study on mutually independent Hamiltonicity of $C_m \times C_n$ and show that IHC$(C_m \times C_n) = 4$, for m is even, n is odd and $|m-n| = 1$.

2 Symbol Definition

Let $Z_n = \{0, 1, 2, ..., n-1\}, n \in N$ and let $V(C_n) = Z_n$. We refer to *toroidal mesh graph* [13] and define graph $C_m \times C_n$, for $m, n \geq 3$ as follows: The vertex set $V(C_m \times C_n) = \{(x,y) : x \in Z_m, y \in Z_n\}$; The edge set $E(C_m \times C_n) = \{((u_1,u_2),(v_1,v_2)) : (u_1,u_2),(v_1,v_2) \in V(C_m \times C_n)$ and $u_1 = v_1, |u_2 - v_2| \in \{1, n-1\}$ or $u_2 = v_2, |u_1 - v_1| \in \{1, m-1\}\}$. We use this symbol definition to construct $C_6 \times C_5$ as Figure 1. A graph G is a *k-regular* graph if for all $x \in V(G)$, $\deg_G(x) = k$. And a graph is *vertex transitive* if every vertex can be mapped to any other vertex by some automorphism [16]. By definition, $C_m \times C_n$ is a 4-regular graph with mn vertices and it is vertex transitive.

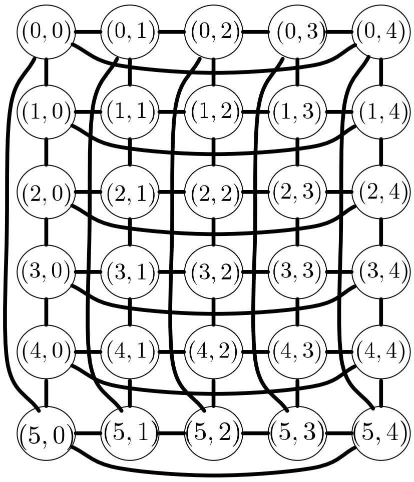

Figure 1: Symbol definition of $C_6 \times C_5$.

In $C_m \times C_n$, we use C^i to denote the *ith subgraph* of $C_m \times C_n$ for $i \in Z_m$, where the vertex set $V(C^i) = \{(i,v) : v \in Z_n\}$; The edge set $E(C^i) = \{((i,v_1),(i,v_2)) : |v_1 - v_2| \in \{1, n-1\}$ and $v_1, v_2 \in Z_n\}$. Then $C_m \times C_n$ can be decomposed into m subgraph C^i for $i \in Z_m$, and each C^i is isomorphic to C_n. Let (i,x) and $(i,y) \in V(C^i)$, then we define $P_{i,x,y,+} = \langle (i,x),(i,x+1),(i,x+2),...,(i,y)\rangle$ is a path of C^i which is from (i,x) to (i,y) and the index of vertices are increased. And $P_{i,x,y,-} = \langle (i,x),(i,x-1),(i,x-2),...,(i,y)\rangle$ is a path of C^i which is from (i,x) to (i,y) and the index of vertices are decreased. Note that the operation $+$ or $-$ should be calculate on Z_n.

3 Main Result

First, we will show Lemma 1 to solve the special case of IHC$(C_m \times C_n)$ for $3 \leq m, n \leq 7$ and $|m-n| = 1$. For the general case, it will be proved in Theorem 1.

Lemma 1 *For any two positive integers $3 \leq m, n \leq 7$, if $|m-n| = 1$ then IHC$(C_m \times C_n) = 4$.*

Proof. Without loss of generality, we assume that m is even, n is odd. Since $C_m \times C_n$ is 4-regular graph, IHC($C_m \times C_n$) ≤ 4 by Property 1. In the follows, we will show that IHC($C_m \times C_n$) ≥ 4 by constructing 4-mutually independent Hamiltonian cycles starting at any vertex of $C_m \times C_n$. We may assume that 4-mutually independent Hamiltonian cycles starting at vertex $(0,0)$ because $C_m \times C_n$ is vertex transitive.

First, we construct four Hamiltonian cycles of $C_4 \times C_3$ starting at $(0,0)$ as Table 1. It is clearly that H_1, H_2, H_3 and H_4 are mutually independent. Since IHC($C_4 \times C_3$) ≥ 4 and $C_4 \times C_3$ is vertex transitive. Hence, IHC($C_4 \times C_3$) $= 4$ can be concluded.

Similarly, we construct four Hamiltonian cycles of $C_4 \times C_5$, $C_6 \times C_5$ and $C_6 \times C_7$ starting at $(0,0)$ as Table 2, 3 and 4, respectively. For the same reasons, IHC($C_m \times C_n$) $= 4$ for these cases can be concluded. It complete this prooof. $\square$

It is not too difficult to see that the following properties are true. These two properties will used to solve the proof of Theorem 1 in some cases.

Property 2 *For any graph G, if there exists two paths P_1, P_2 and a subgraph $C_n \subseteq G$ with $C_n = \langle x_0, x_1, ..., x_{n-1}, x_0\rangle$, $P_1(t_1 + j) = x_{u+j}, P_2(t_2 + j) = x_{v+j}$ for $0 \leq j \leq n-1$ and $v - u = a \geq 0, t_1 - t_2 = b$. If any one of the following condition holds, then these two paths are not at the same vertex of C_n in the same time.*
(1) If $a = 0$ and $b \neq 0$. (2) If $a \neq 0$ and ($b \neq -a$ or $n - a$).

Table 5 is an illustration of Property 2 where $P_1 = \langle x_0, x_1, ..., x_{n-1}\rangle, P_2 = \langle x_1, x_2, ..., x_{n-1}, x_0\rangle$ and $a = 1$.

Property 3 *For any graph G, if there exists two paths P_1, P_2 and a subgraph $C_n \subseteq G$ with $V(C_n) = \langle x_0, x_1, ..., x_{n-1}, x_0\rangle$, $P_1(t_1 + j) = x_{u-j}, P_2(t_2 + j) = x_{v-j}$ for $0 \leq j \leq n-1$ and $v - u = a \geq 0, t_1 - t_2 = b$. If any one of the following condition holds, then these two paths are not at the same vertex of C_n in the same time.*
(1) If $a = 0$ and $b \neq 0$. (2) If $a \neq 0$ and ($b \neq a$ or $a - n$).

Table 6 is an illustration of Property 3 where $P_1 = \langle x_0, x_{n-1}, x_{n-2}, ..., x_1\rangle$, $P_2 = \langle x_1, x_0, x_{n-1}, ..., x_2\rangle$ and $a = 1$.

In the follows, we will show that IHC($C_m \times C_n$) $= 4$ for $m, n \geq 3$ and $|m - n| = 1$.

Theorem 1 *For two positive integers $m, n \geq 3$, if $|m - n| = 1$ then IHC($C_m \times C_n$) $= 4$.*

Proof. Without loss of generality, we assume that m is even, n is odd. For $C_4 \times C_3$, $C_4 \times C_5$, $C_6 \times C_5$ and $C_6 \times C_7$, by Lemma 1, it is proved. Now, we consider about $m \geq 8$ and $n \geq 7$. Since $C_m \times C_n$ is 4-regular graph, IHC($C_m \times C_n$) ≤ 4 by Property 1. Next, will show that IHC($C_m \times C_n$) ≥ 4 by constructing 4-mutually independent Hamiltonian cycles starting at any vertex of $C_m \times C_n$. We may assume that 4-mutually independent Hamiltonian cycles starting at vertex $e = (0,0)$ because $C_m \times C_n$ is vertex transitive. As we wish, we construct four Hamiltonian cycles starting at vertex e that they do not pass the same subgraph in most of the time as follows.

$H_1 = \langle (0,0), (0,1), (1,1), P_{1,1,2,-}, (1,2), (0,2), P_{0,2,n-1,+}, (0,n-1), (m-1,n-1), (m-2,n-1), ..., (3,n-1), (2,n-1), P_{2,n-1,n-2,+}, (2,n-2), (3,n-2), P_{3,n-2,0,-}, (3,0), (4,0), P_{4,0,n-2,+}, (4,n-2), (5,n-2), P_{5,n-2,0,-}, (5,0), ..., (m-1,n-2), P_{m-1,n-2,0,-}, (m-1,0), (0,0)\rangle$,

$H_2 = \langle (0,0), (m-1,0), P_{m-1,0,n-2,+}, (m-1,n-2), (0,n-2), P_{0,n-2,1,-}, (0,1), (1,1), P_{1,1,0,+}, (1,0), (2,0), P_{2,0,n-1,+}, (2,n-1), (3,n-1), P_{3,n-1,0,-}, (3,0), (4,0), P_{4,0,n-1,+}, (4,n-1), ..., (m-2,0), P_{m-2,0,n-1,+}, (m-2,n-1), (m-1,n-1), (0,n-1), (0,0)\rangle$,

$H_3 = \langle (0,0), (1,0), P_{1,0,1,-}, (1,1), (2,1), P_{2,1,0,+}, (2,0), (3,0), P_{3,0,1,-}, (3,1), ..., (m-2,1), P_{m-2,1,0,+}, (m-2,0), (m-1,0), P_{m-1,0,n-1,+}, (m-1,n-1), (0,n-1), P_{0,n-1,0,-}, (0,0)\rangle$,

$H_4 = \langle (0,0), (0,n-1), (m-1,n-1), P_{m-1,n-1,n-2,+}, (m-1,n-2), (m-2,n-2), P_{m-2,n-2,n-3,+}, (m-2,n-3), (m-3,n-3), P_{m-3,n-3,n-2,-}, (m-3,n-2), (m-4,n-2), P_{m-4,n-2,n-3,+}, (m-4,n-3), (m-5,n-3), P_{m-5,n-3,n-2,-}, (m-5,n-2), ..., (2,n-2), P_{2,n-2,n-3,+}, (2,n-3), (1,n-3), P_{1,n-3,1,-}, (1,1), (0,1), P_{0,1,n-2,+}, (0,n-2), (1,n-2), (1,n-1), (1,0), (0,0)\rangle$.

According to the construction of four Hamiltonian cycles, we know that H_1 and H_2 belong to the same subgraph in most of the time. So, we divide this proof into two cases

Case 1 : H_1 and H_2.

According to H_1 and H_2 belong to the same subgraph C^j for some $j \in Z_n$ or not, we have following subcases. Figure 2 is the illustration for these three subcases on $C_8 \times C_7$.

Case 1.1 : $t = 2, mn-1$ or mn.

By the construction of H_1 and H_2, we know that $H_1(2) = (0,1), H_1(mn-1) = (m-1,1), H_1(mn) = (m-1,0)$ and $H_2(2) = (m-1,0), H_2(mn-1) = (m-1,n-1), H_2(mn) = (0,n-1)$. It is clearly that H_1 and H_2 are in-

t	1	2	3	4	5	6	7	8	9	10	11	12	13
H_1	$(0,0)$	$(0,1)$	$(1,1)$	$(1,0)$	$(1,2)$	$(2,2)$	$(2,0)$	$(2,1)$	$(3,1)$	$(3,0)$	$(3,2)$	$(0,2)$	$(0,0)$
H_2	$(0,0)$	$(3,0)$	$(3,1)$	$(3,2)$	$(2,2)$	$(2,0)$	$(2,1)$	$(1,1)$	$(0,1)$	$(0,2)$	$(1,2)$	$(1,0)$	$(0,0)$
H_3	$(0,0)$	$(1,0)$	$(2,0)$	$(3,0)$	$(3,1)$	$(2,1)$	$(1,1)$	$(1,2)$	$(2,2)$	$(3,2)$	$(0,2)$	$(0,1)$	$(0,0)$
H_4	$(0,0)$	$(0,2)$	$(0,1)$	$(1,1)$	$(2,1)$	$(3,1)$	$(3,2)$	$(2,2)$	$(1,2)$	$(1,0)$	$(2,0)$	$(3,0)$	$(0,0)$

Table 1: 4-mutually independent Hamiltonian cycles in $C_4 \times C_3$ starting at $(0,0)$.

t	1	2	3	4	5	6	7	8	9	10	11	12	13
H_1	$(0,0)$	$(0,1)$	$(0,2)$	$(0,3)$	$(0,4)$	$(1,4)$	$(1,0)$	$(1,1)$	$(1,2)$	$(1,3)$	$(2,3)$	$(3,3)$	$(3,4)$
H_2	$(0,0)$	$(3,0)$	$(3,4)$	$(3,3)$	$(3,2)$	$(3,1)$	$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$	$(2,0)$	$(1,0)$	$(1,4)$
H_3	$(0,0)$	$(1,0)$	$(2,0)$	$(3,0)$	$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$	$(2,4)$	$(2,3)$	$(2,2)$	$(2,1)$	$(1,1)$
H_4	$(0,0)$	$(0,4)$	$(1,4)$	$(2,4)$	$(2,3)$	$(2,2)$	$(1,2)$	$(1,3)$	$(0,3)$	$(0,2)$	$(0,1)$	$(1,1)$	$(2,1)$

t	14	15	16	17	18	19	20	21
H_1	$(2,4)$	$(2,0)$	$(2,1)$	$(2,2)$	$(3,2)$	$(3,1)$	$(3,0)$	$(0,0)$
H_2	$(1,3)$	$(1,2)$	$(1,1)$	$(0,1)$	$(0,2)$	$(0,3)$	$(0,4)$	$(0,0)$
H_3	$(1,2)$	$(1,3)$	$(1,4)$	$(0,4)$	$(0,3)$	$(0,2)$	$(0,1)$	$(0,0)$
H_4	$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$	$(3,0)$	$(2,0)$	$(1,0)$	$(0,0)$

Table 2: 4-mutually independent Hamiltonian cycles in $C_4 \times C_5$ starting at $(0,0)$.

t	1	2	3	4	5	6	7	8	9	10	11	12	13
H_1	$(0,0)$	$(0,1)$	$(0,2)$	$(0,3)$	$(0,4)$	$(1,4)$	$(1,0)$	$(1,1)$	$(1,2)$	$(1,3)$	$(2,3)$	$(2,2)$	$(2,1)$
H_2	$(0,0)$	$(5,0)$	$(5,4)$	$(5,3)$	$(5,2)$	$(5,1)$	$(4,1)$	$(4,2)$	$(4,3)$	$(4,4)$	$(4,0)$	$(3,0)$	$(3,4)$
H_3	$(0,0)$	$(1,0)$	$(2,0)$	$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$	$(3,4)$	$(3,0)$	$(3,1)$	$(3,2)$	$(3,3)$	$(4,3)$
H_4	$(0,0)$	$(0,4)$	$(0,3)$	$(0,2)$	$(0,1)$	$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$	$(2,4)$	$(3,4)$	$(4,4)$	$(5,4)$

t	14	15	16	17	18	19	20	21	22	23	24	25	26
H_1	$(2,0)$	$(2,4)$	$(3,4)$	$(3,0)$	$(3,1)$	$(3,2)$	$(3,3)$	$(4,3)$	$(4,2)$	$(4,1)$	$(4,0)$	$(4,4)$	$(5,4)$
H_2	$(3,3)$	$(3,2)$	$(3,1)$	$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$	$(2,0)$	$(1,0)$	$(1,4)$	$(1,3)$	$(1,2)$	$(1,1)$
H_3	$(5,3)$	$(5,4)$	$(4,4)$	$(4,0)$	$(5,0)$	$(5,1)$	$(4,1)$	$(4,2)$	$(5,2)$	$(0,2)$	$(0,3)$	$(0,4)$	$(1,4)$
H_4	$(5,0)$	$(5,1)$	$(5,2)$	$(5,3)$	$(4,3)$	$(3,3)$	$(2,3)$	$(2,2)$	$(2,1)$	$(3,1)$	$(3,2)$	$(4,2)$	$(4,1)$

t	27	28	29	30	31
H_1	$(5,3)$	$(5,2)$	$(5,1)$	$(5,0)$	$(0,0)$
H_2	$(0,1)$	$(0,2)$	$(0,3)$	$(0,4)$	$(0,0)$
H_3	$(1,3)$	$(1,2)$	$(1,1)$	$(0,1)$	$(0,0)$
H_4	$(4,0)$	$(3,0)$	$(2,0)$	$(1,0)$	$(0,0)$

Table 3: 4-mutually independent Hamiltonian cycles in $C_6 \times C_5$ starting at $(0,0)$.

dependent because $n \geq 7$ in this subcase.

Case 1.2 : $3 \leq t \leq 3n+m-3$.

For $3 \leq t \leq n+2$, $H_1(t) \in C^1$. For $3 \leq t \leq n$, $H_2(t) \in C^{m-1}$ and for $n+1 \leq t \leq n+2$, $H_2(t) \in C^0$. Since H_1 and H_2 belong to different subgraphs, H_1 and H_2 are independent for $3 \leq t \leq n+2$.

For $n+3 \leq t \leq 2n$, $H_1(t) \in C^0$. And for $n+1 \leq t \leq 2n-2$, $H_2(t) \in C^0$. By the construction of H_1 and H_2, we know that $H_1(n+k) = (0, k-1)$ and $H_2(n+k) = (0, n-k-1)$ for $3 \leq k \leq n-2$. Since n is odd and $k-1 \neq n-k-1$, we know that these two cycles are independent for $n \geq 7$. Table 7 is an illustration of this subcase.

For $t = 2n+i$ where $1 \leq i \leq m-3$, $H_1(t) = (m-i, n-1)$. And for $2n-1 \leq t \leq 2n+m-3$, $H_2(t) \in C^1$. It is clearly that H_1 and H_2 are independent.

For $2n+m-2 \leq t \leq 3n+m-3$, $H_1(t)$ and $H_2(t) \in C^2$. Two cycles H_1 and H_2 go the same direction in C^2. By the construction of H_1 and H_2, we know that $H_1(2n+m-2) = (2, n-1)$, $H_2(2n+m-2) = (2, 0)$. By Property 2(2)(where $a = (n-1)-0 \neq 0$ and $b = (2n+m-2)-(2n+m-2) = 0$ in this case), H_1 and H_2 are independent for $2n+m-2 \leq t \leq 3n+m-3$.

Case 1.3 : $3n+m-2 \leq t \leq mn-2$.

For $1 \leq i \leq m-4$, $3n+m-2+(i-1)(n-1) \leq t_1 \leq 3n+m-3+i(n-1)$ $H_1(t_1) \in C^{i+2}$. For $1 \leq j \leq m-5$, $3n+m-2+(j-1)n \leq t_2 \leq 3n+m-3+jn$, $H_2(t_2) \in C^{j+2}$. For all $i = j$, two cycles H_1 and H_2 go the same direction in C^{i+2}. By the construction of H_1 and H_2, we know that if i and j are odd $H_1(3n+m-2+(i-1)(n-1)) = (i+2, n-2)$, $H_2(3n+m-2+(j-1)n) = (j+2, n-1)$. And if i and j are even $H_1(3n+m-2+(i-1)(n-1)) =$

t	1	2	3	4	5	6	7	8	9	10	11	12	13
H_1	$(0,0)$	$(0,1)$	$(0,2)$	$(0,3)$	$(0,4)$	$(0,5)$	$(0,6)$	$(1,6)$	$(1,0)$	$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$
H_2	$(0,0)$	$(5,0)$	$(5,6)$	$(5,5)$	$(5,4)$	$(5,3)$	$(5,2)$	$(5,1)$	$(4,1)$	$(4,2)$	$(4,3)$	$(4,4)$	$(4,5)$
H_3	$(0,0)$	$(1,0)$	$(2,0)$	$(3,0)$	$(4,0)$	$(5,0)$	$(5,1)$	$(5,2)$	$(5,3)$	$(5,4)$	$(5,5)$	$(5,6)$	$(4,6)$
H_4	$(0,0)$	$(0,6)$	$(0,5)$	$(0,4)$	$(0,3)$	$(0,2)$	$(0,1)$	$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$	$(1,5)$	$(1,6)$

t	14	15	16	17	18	19	20	21	22	23	24	25	26
H_1	$(1,5)$	$(2,5)$	$(2,6)$	$(2,0)$	$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$	$(3,4)$	$(3,3)$	$(3,2)$	$(3,1)$	$(3,0)$
H_2	$(4,6)$	$(4,0)$	$(3,0)$	$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$	$(3,5)$	$(3,6)$	$(2,6)$	$(2,5)$	$(2,4)$	$(2,3)$
H_3	$(3,6)$	$(3,5)$	$(4,5)$	$(4,4)$	$(4,3)$	$(4,2)$	$(4,1)$	$(3,1)$	$(3,2)$	$(2,2)$	$(2,1)$	$(1,1)$	$(1,2)$
H_4	$(2,6)$	$(3,6)$	$(4,6)$	$(5,6)$	$(5,0)$	$(5,1)$	$(5,2)$	$(5,3)$	$(5,4)$	$(5,5)$	$(4,5)$	$(4,4)$	$(4,3)$

t	27	28	29	30	31	32	33	34	35	36	37	38	39
H_1	$(3,6)$	$(3,5)$	$(4,5)$	$(4,4)$	$(4,3)$	$(4,2)$	$(4,1)$	$(4,0)$	$(4,6)$	$(5,6)$	$(5,5)$	$(5,4)$	$(5,3)$
H_2	$(2,2)$	$(2,1)$	$(2,0)$	$(1,0)$	$(1,6)$	$(1,5)$	$(1,4)$	$(1,3)$	$(1,2)$	$(1,1)$	$(0,1)$	$(0,2)$	$(0,3)$
H_3	$(1,3)$	$(2,3)$	$(3,3)$	$(3,4)$	$(2,4)$	$(1,4)$	$(1,5)$	$(2,5)$	$(2,6)$	$(1,6)$	$(0,6)$	$(0,5)$	$(0,4)$
H_4	$(4,2)$	$(4,1)$	$(4,0)$	$(3,0)$	$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$	$(3,5)$	$(2,5)$	$(2,4)$	$(2,3)$	$(2,2)$

t	40	41	42	43
H_1	$(5,2)$	$(5,1)$	$(5,0)$	$(0,0)$
H_2	$(0,4)$	$(0,5)$	$(0,6)$	$(0,0)$
H_3	$(0,3)$	$(0,2)$	$(0,1)$	$(0,0)$
H_4	$(2,1)$	$(2,0)$	$(1,0)$	$(0,0)$

Table 4: 4-mutually independent Hamiltonian cycles in $C_6 \times C_7$ starting at $(0,0)$.

t	t_1-n+1	$t_1-...$	t_1-2	t_1-1	t_1	t_1+1	t_1+2	$t_1+...$	t_1+n-1	t_1-n
P_1					x_0	x_1	x_2	...	x_{n-1}	
$b=-1, P_2$						x_1	x_2	...	x_{n-1}	x_0
$b=n-1, P_2$	x_1	...	x_{n-2}	x_{n-1}	x_0					
$b=0, P_2$					x_1	x_2	x_3	...	x_0	
$b=1, P_2$				x_1	x_2	x_3	x_4	...		

Table 5: A illustration of Property 2 where $a=1, b=-1, n-1, 0$ or 1.

t	t_1-1	t_1	t_1+1	t_1+2	$t_1+...$	t_1+n-1	t_1+n	t_1+n+1	$t_1+...$	t_1+2n-2
P_1		x_0	x_{n-1}	x_{n-2}	...	x_1				
$b=1, P_2$	x_1	x_0	x_{n-1}	x_{n-2}	...					
$b=1-n, P_2$						x_1	x_0	x_{n-1}	...	x_2
$b=0, P_2$		x_1	x_0	x_{n-1}	...	x_2				
$b=-1, P_2$			x_1	x_0	...	x_3	x_2			

Table 6: A illustration of Property 3 where $a=1, b=1, 1-n, 0$ or -1.

$(i+2,0)$, $H_2(3n+m-2+(j-1)n)=(j+2,0)$. If $i=j$, i is odd and $[(3n+m-2)+(i-1)n]-[(3n+m-2)+(j-1)(n-1)]=i-1 \neq [(n-1)-(n-2)]$ or $[(n-1)-(n-2)]-n$ for $n \geq 7$ and i is odd, then H_1 and H_2 are independent by Property 3(2). Similarly, if $i=j$, i is even, then H_1 and H_2 are independent by Property 2(1).

According to the construction of H_1 and H_2, we know that for $3n+m-2+(m-4)(n-1)=mn-(n-2) \leq t_3 \leq mn-2$, $H_1(t_3) \in C^{m-1}$ and $H_2(t_3) \in C^{m-2}$. It is clearly that H_1 and H_2 are independent. Hence, H_1 and H_2 are independent in this subcase.

Case 2 : H_1, H_3, H_4 and H_2, H_3, H_4.

Note that H_1 and H_2 are independent, proved by Case 1. Now, we consider $H_i(t), 2 \leq t \leq mn, i \in \{1,3,4\}$ and $H_j(t), 2 \leq t \leq mn, j \in \{2,3,4\}$, if these three Hamiltonian cycles are belong to different subgraph, they are mutually independent intuitively. Therefore, we only need to discuss the following subcases. Figure 3 is an illustration for these six subcases on $C_8 \times C_7$.

Case 2.1 : $t=2, mn-1$ or mn.

By the construction of H_1, H_2, H_3 and H_4, we know that $H_1(2)=(0,1), H_1(mn-1)=(m-1,1), H_1(mn)=(m-1,0)$; $H_2(2)=(m-1,0), H_2(mn-1)=(m-1,n-1), H_2(mn)=(0,n-1)$; $H_3(2)=(1,0), H_3(mn-1)=(0,2), H_3(mn)=(0,1)$ and $H_4(2)=(0,n-1), H_4(mn-1)=(1,n-1), H_4(mn)=(1,0)$. It is clearly that H_1, H_2, H_3 and H_4 are mutually independent because $n \geq 7$ in this subcase.

Case 2.2 : $3 \le t \le n+1$.

For $3 \le t_1 \le n+2$, $H_1(t_1) \in C^1$, and for $2 \le t_2 \le n+1$, $H_3(t_2) \in C^1$. Two cycles H_1 and H_3 go the same direction in C^1. By the construction of H_1 and H_3, we know that $H_1(3) = (1,1), H_3(2) = (1,0)$. Since $2-3 = -1 \ne 1-0$ or $(1-0)-n$ for $n \ge 7$, H_1 and H_3 are independent by Property 3(2).

Similarly, for $2 \le t_3 \le n$, $H_2(t_3) \in C^{m-1}$, and for $3 \le t_4 \le n+2$, $H_4(t_2) \in C^{m-1}$. Two cycles H_2 and H_4 go the same direction in C^{m-1}. By the construction of H_2 and H_4, we know that $H_2(2) = (m-1,0), H_4(3) = (m-1,n-1)$. Since $(n-1)-0 = n-1 \ne -(2-3)$ or $n-(2-3)$, H_2 and H_4 are independent by Property 2(2).

Case 2.3 : $t = 2n+3$ or $2n+m-3$.

By the construction of H_1, H_3 and H_4, we know that $H_1(2n+3) = (m-3,n-1); H_4(2n+3) = (m-3,n-3)$, and $H_1(2n+m-3) = (3,n-1); H_3(2n+m-3) = (3,n-m+5)$. Because $n-1 \ne n-3$ and $n-1 \ne n-m+5$ for $m \ge 8$, H_1, H_3 and H_4 are mutually independent in this subcase.

Case 2.4 : $n((m/2)-1)+2 \le t \le n(m/2)+2$.

For $n((m/2)-1)+2 \le t_1 \le n((m/2)-1)+n+1$, $H_3(t_1) \in C^{m/2}$, and for $n((m/2)-1)+3 \le t_2 \le n((m/2)-1)+n+2 = n(m/2)+2$, $H_4(t_2) \in C^{m/2}$. Two cycles H_3 and H_4 go the same direction in $C^{m/2}$. By the construction of H_3 and H_4, we know that if m mod $4=0$, then $H_3(n((m/2)-1)+2) = (m/2,1), H_4(n((m/2)-1)+3) = (m/2,n-2)$. And if m mod $4=2$, then $H_3(n((m/2)-1)+2) = (m/2,0), H_4(n((m/2)-1)+3) = (m/2,n-3)$. Since $[n((m/2)-1)+2]-[n((m/2)-1)+3] = -1 \ne -(n-2-1)$ or $n-(n-2-1)$ for $n \ge 7$ when m mod $4=0$; $[n((m/2)-1)+2]-[n((m/2)-1)+3] = -1 \ne [(n-3)-0]$ or $[(n-3)-0]-n$ for $n \ge 7$ when m mod $4=2$, H_3 and H_4 are independent by Properties 2(2) and 3(2) in this subcase.

Case 2.5 : $n(m/2)+3 \le t \le n((m/2)+1)+2$.

For $n(m/2)+(m/2)-n+2 \le t_1 \le n(m/2)+(m/2)$, $H_1(t_1) \in C^{(m/2)-1}$, and for $n(m/2)+3 \le t_2 \le n(m/2)+n+2$, $H_4(t_2) \in C^{(m/2)-1}$. Two cycles H_1 and H_4 go the same direction in $C^{(m/2)-1}$. By the construction of H_1 and H_4, we know that if m mod $4=0$, then $H_1(n(m/2)+(m/2)-n+2) = ((m/2)-1,n-2), H_4(n(m/2)+3) = ((m/2)-1,n-3)$. And if m mod $4=2$, then $H_1(n(m/2)+(m/2)-n+2) = ((m/2)-1,0), H_4(n(m/2)+3) = ((m/2)-1,n-2)$. Since $[n(m/2)+3]-[n(m/2)+(m/2)-n+2] = -(n-(m/2))+1 \ne 1$ or $1-n$ for n is odd and m mod $4=0$; $[n(m/2)+(m/2)-n+2]-[n(m/2)+3] = (n-(m/2))-1 \ne -[(n-2)-0]$ or $n-[(n-2)-0]$ for $|m-n|=1$ and m mod $4=2$, H_1 and H_4 are independent by Properties 3(2) and 2(2).

Similarly, for $n(m/2)-1 \le t_3 \le n(m/2)+n-2$, $H_2(t_3) \in C^{(m/2)-1}$, and for $n(m/2)+3 \le t_2 \le n(m/2)+n+2$, $H_4(t_2) \in C^{(m/2)-1}$. Two cycles H_2 and H_4 go the same direction in $C^{(m/2)-1}$. By the construction of H_2 and H_4, we know that if m mod $4=0$, then $H_2(n(m/2)-1) = ((m/2)-1,n-1), H_4(n(m/2)+3) = ((m/2)-1,n-3)$. And if m mod $4=2$, then $H_2(n(m/2)-1) = ((m/2)-1,0), H_4(n(m/2)+3) = ((m/2)-1,n-2)$. Since $[n(m/2)+3]-[n(m/2)-1] = 4 \ne [(n-1)-(n-3)]$ or $[(n-1)-(n-3)]-n$ for $n \ge 7$ and m mod $4=0$; $[n(m/2)-1]-[n(m/2)+3] = -4 \ne -[(n-2)-0]$ or $n-[(n-2)-0]$ for $n \ge 7$ and m mod $4=2$, H_2 and H_4 are independent by Property 3(2) and 2(2).

Case 2.6 : $mn-n+2 \le t \le mn-3$.

For $mn-n+2 \le t_1 \le mn$, $H_3(t_1) \in C^0$. And for $mn-n \le t_2 \le mn-3$, $H_4(t_2) \in C^0$. By the construction of H_3 and H_4, we know that $H_3(mn-k) = (0,k+1)$ and $H_4(mn-k) = (0,n-k+1)$ for $3 \le k \le 7$. Since n is odd and $k+1 \ne n-k+1$, we know that these two cycles are independent for $n \ge 7$. Table 8 is an illustration of this subcase.

From the above two cases, we can know that H_1, H_2, H_3 and H_4 starting at vertex e are mutually independent and IHC$(C_m \times C_n) \ge 4$ because $C_m \times C_n$ is vertex transitive. Hence, IHC$(C_m \times C_n) = 4$ can be concluded. □

We construct 4-mutually independent Hamiltonian cycles for $C_8 \times C_7$ in Figure 4, 5, 6, 7 as an example of Theorem 1.

4 Conclusion

In this paper, we discuss the mutually independent Hamiltonian cycles of $C_k \times C_{k+1}$, for any positive integer $k \ge 3$, and get the optimal result on it. Note that a Hamilton cycles in any graph G is a cycle C_k with $k = |V(G)|$. So, if two graphs G_1 and G_2 are Hamiltonian with $|V(G_1)| = k$, $|V(G_2)| = k+1$ and $k \ge 3$, then graph $C_k \times C_{k+1}$ will be a spanning subgraph of $G_1 \times G_2$. Hence, we have the following corollary.

Corollary 1 *For any graphs G_1, G_2, if G_1 and G_2 are Hamiltonian and $|V(G_2)|-1 = |V(G_1)| \ge 3$, then IHC$(G_1 \times G_2) \ge 4$.*

In the future, discussing the mutually independent Hamiltonian cycles of $C_m \times C_n$, for $m, n \ge$

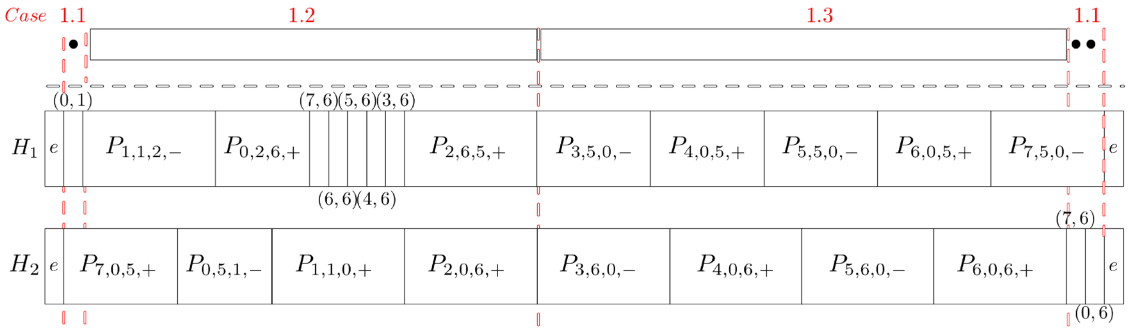

Figure 2: The construction of two Hamiltonian cycles of $C_8 \times C_7$ start at e

t	$n+3$	$n+4$	$n+5$	...	$n+k$	...	$2n-2$	$2n-1$	$2n$
H_1	$(0,2)$	$(0,3)$	$(0,4)$	...	$(0,k-1)$	...	$(0,n-3)$	$(0,n-2)$	$(0,n-1)$
H_2	$(0,n-4)$	$(0,n-5)$	$(0,n-6)$	...	$(0,n-k-1)$	...	$(0,1)$	$(1,1)$	$(1,2)$

Table 7: A illustration of H_1 and H_2.

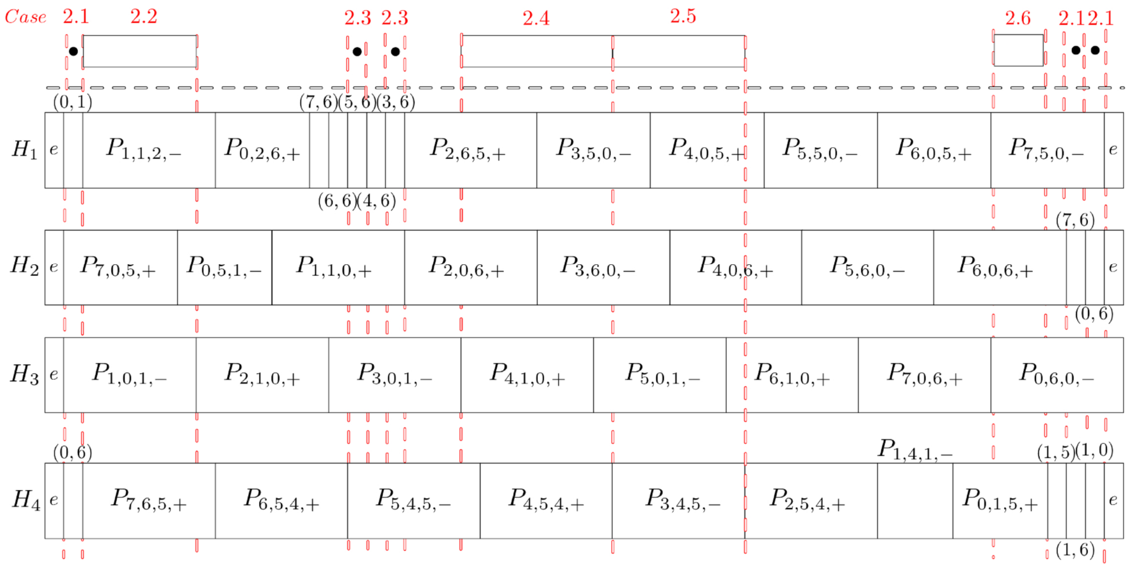

Figure 3: The construction of four Hamiltonian cycles of $C_8 \times C_7$ start at e

t	$mn-n$	$mn-n+1$	$mn-n+2$	$mn-n+3$	...	$mn-k$	...	$mn-3$
H_3	$(m-1,n-2)$	$(m-1,n-1)$	$(0,n-1)$	$(0,n-2)$	...	$(0,k+1)$	...	$(0,4)$
H_4	$(0,1)$	$(0,2)$	$(0,3)$	$(0,4)$	...	$(0,n-k+1)$	...	$(0,n-2)$

Table 8: A illustration of H_3 and H_4.

4 and m, n are even is an interesting question. Furthermore, we can discuss IHC(G) for $G = C_{n_1} \times C_{n_2} \times ... \times C_{n_k}$, for any positive integer $n_1 \geq n_2 \geq ... \geq n_k \geq 3$ and $k \geq 3$.

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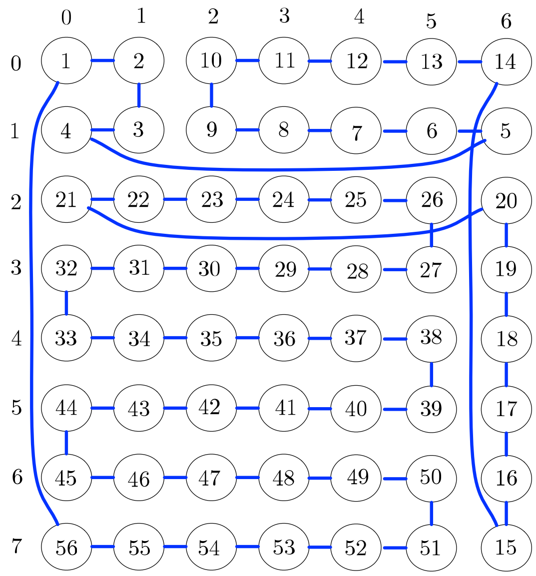

Figure 4: Hamiltonian cycles H_1 in $C_8 \times C_7$.

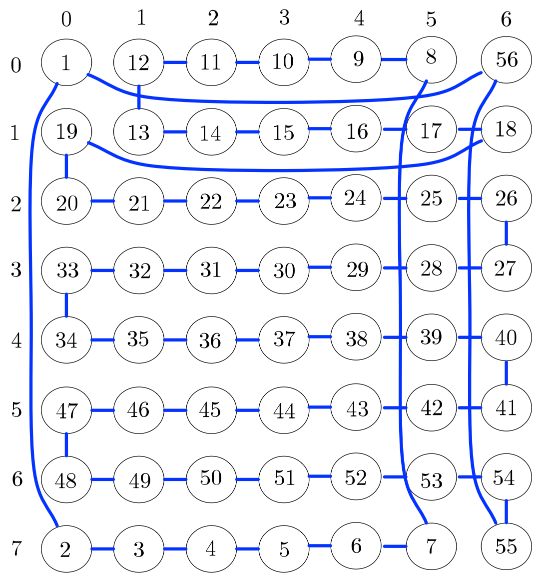

Figure 5: Hamiltonian cycles H_2 in $C_8 \times C_7$.

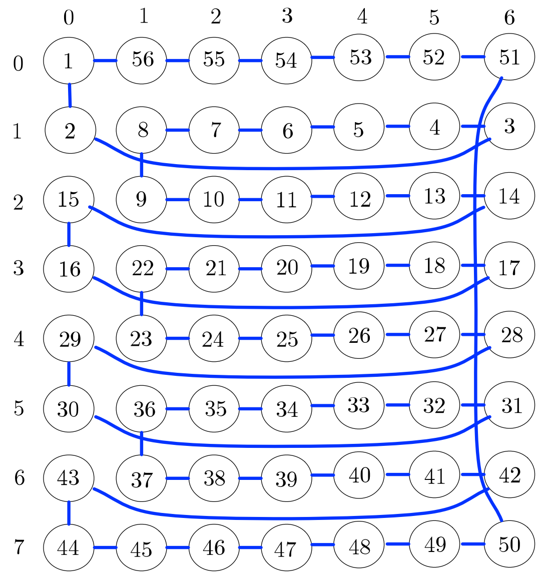

Figure 6: Hamiltonian cycles H_3 in $C_8 \times C_7$.

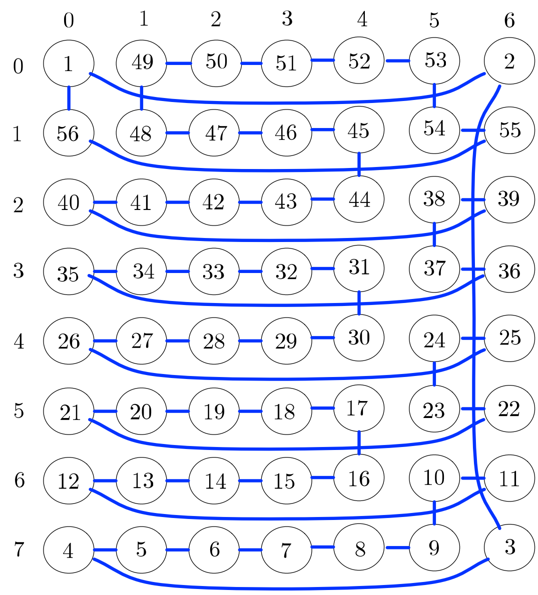

Figure 7: Hamiltonian cycles H_4 in $C_8 \times C_7$.

References

[1] S. G. Akl, "Parallel computation:models and methods." *Prentice Hall*, 1997.

[2] S. Y.-P. Chang, J. S.-T. J., C.-K. Lin, J. J. M. Tan, L.-H. Hsu, "Mutually independent hamiltonian connectivity of (n, k)-star graphs," *Annals of Combinatorics*, Vol 13, pp 27-52, 2009.

[3] G. Chartrand and O. R. Oellermann, "Applied and algorithmic graph theory" *McGraw-Hill*, 1993.

[4] T. L. Kueng, T. Liang, and L. H. Hsu, "Mutually independent hamiltonian cycles of binary wrapped butterfly graphs," *Mathematical and Computer Modelling,* Vol. 48, no. 11-12, pp. 1814-1825, 2008.

[5] T. L. Kung, C. K. Lin, T. Liang, J. J. M. Tan, and L. H. Hsu, "Fault free mutually independent hamiltonian cycles of faulty star graphs,"

International Journal of Computer Mathematics, Vol. 88, no. 4, pp. 731-746, 2011.

[6] F. T. Leighton, "Introduction to Parallel Algorithms and Architectures: Arrays, Trees, Hypercubes" *Morgan Kaufmann*, 1992.

[7] Y. L. Lai, D. C. Yu, and L. H. Hsu, "Mutually independent hamiltonian cycle of burnt pancake graphs," *IEICE-Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, Vol. E94A, no. 7, pp. 1553-1557, 2011.

[8] C.-D. Lin, "Mutually independent hamiltonian cycles on arrangement graphs" *Masters thesis*, , Chung Yuan Christian University, 2011.

[9] C. K. Lin, J. J. M. Tan, H. M. Huang, D. F. Hsu, and L. H. Hsu, "Mutually independent hamiltonian cycles for the pancake graphs and the star graphs," *Discrete Mathematics*, Vol. 309, no. 17, pp. 5474-5483, 2009.

[10] Y. K. Shih, H. C. Chuang, S. S. Kao, and J. J. M. Tan, "Mutually independent hamiltonian cycles in dual-cubes," *Journal of Supercomputing*, Vol. 54, no. 2, pp. 239-251, 2010.

[11] H. Su, J. L. Pan, and S. S. Kao, "Mutually independent hamiltonian cycles in k-ary n-cubes when k is even," *Computers and Electrical Engineering*, Vol. 37, no. 3, pp. 319-331, 2011.

[12] C. M. Sun, C. K. Lin, H. M. Huang, and L. H. Hsu, "Mutually independent Hamiltonian paths and cycles in hypercubes," *J Interconnect Networks,* Vol 7, pp. 235-255, 2006.

[13] K. W. Tang and S. A. Padubidri, "Diagonal and toroidal mesh networks" *Computers, IEEE Transactions on,* Vol. 43, no. 7, pp. 815-826, 1994.

[14] K.-S. Wu and J. S.-T. Juan, "Mutually independent hamiltonian cycles of $C_n \times C_n$," *Proceedings of ICCSSE 2012 : International Conference on Computer Science and Software Engineering*, 2011.

[15] K.-S. Wu and J. S.-T. Juan, "Mutually Independent Hamiltonian Cycles of $C_m \times C_n$ when m, n Are Odd" *Proceedings of the 29th Workshop on Combinatorial Mathematics and Computation Theory*, pp 165-170, 2012

[16] J. M. Xu, "Topological structure and analysis of interconnection networks" *Kluwer Academic*, 2001.

[17] J. M. Xu and M. Ma, "Survey on path and cycle embedding in some networks," *Frontiers of Mathematics in China*, Vol. 4, no. 2, pp. 217-252, 2007.